

Supplemental Material for *Quantum melting a Wigner crystal into Hall liquids*

Aidan P. Reddy* and Liang Fu

Department of Physics, Massachusetts Institute of Technology, Cambridge, Massachusetts 02139, USA

(Dated: February 13, 2026)

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I. EXTENDED RESULTS

A. Phase diagrams at $\nu = 2$ and $B = 0$

In Fig. 1, we show variational energies at $\nu = 2$ and at $B = 0$. The data at $\nu = 2$ shown in Fig. 1(a) indicate a quantum Hall to Wigner crystal transition near $r_s \approx 38$. The data at $B = 0$ in Fig. 1(c) indicate a Fermi liquid to Wigner crystal transition near $r_s \approx 33$. We have compared energies from the integer quantum Hall and Landau level basis expansion (or, at $B = 0$, plane wave expansion) ansätze described in the main text with Gaussian orbitals, which we explain in Sec. III D. We find that the variational wavefunctions with Gaussian orbitals have energies close to those with Landau level or plane wave basis expanded orbitals.

Our estimate of the critical r_s at $B = 0$ is in agreement with the value $r_s \approx 33$ implied by Fig. 6 of Ref. [1], but is somewhat higher than $r_s \approx 27$ implied by Fig. 1 of the earlier work Ref. [2]. Both Refs. [1, 2] used diffusion Monte Carlo. However, Ref. [1] obtained a higher estimate of the critical r_s than Ref. [2] because Ref. [1] found a lower energy for the fully spin-polarized Fermi liquid phase than Ref. [2] and presented this alongside data for the fully spin-polarized Wigner crystal taken from Ref. [2].

The purpose of our $B = 0$ calculation is merely to substantiate our conclusion that the liquid-crystal transition tends toward higher r_s as B increases (restricting to integral ν), not to obtain state-of-the-art accuracy. Thus, for simplicity, we do not twist-average so finite size effects at $B = 0$ may be significant [3]. As we explain in Section III A, twist-averaging is entirely unnecessary when $B \neq 0$.

* areddy@mit.edu

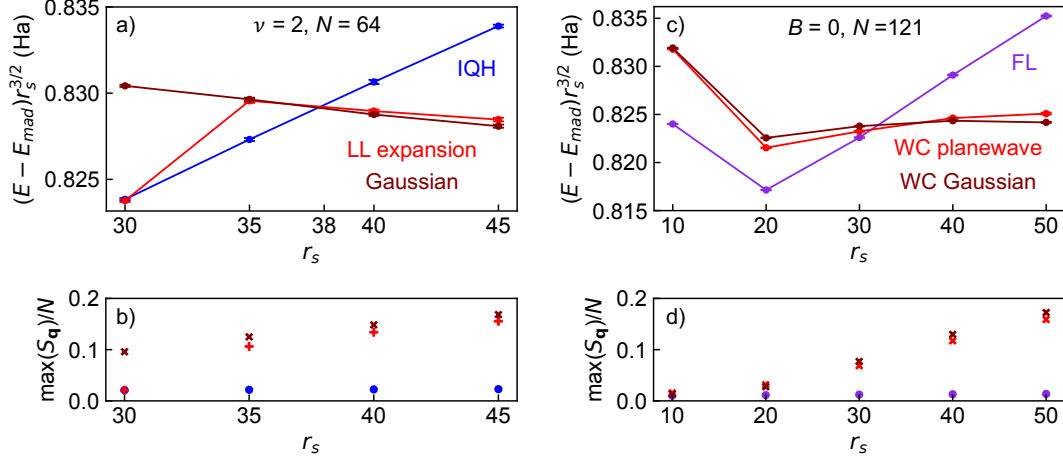


FIG. 1. **Phase diagrams at $\nu = 2$ and $B = 0$.** Variational energies (a,c) and maxima of the static structure factor S_q (b,d) at $\nu = 2$ (a,b) and $B = 0$ (c,d). At $\nu = 2$, $r_s = 30$ the Landau level expansion wavefunction flows through optimization to a liquid wavefunction that is essentially identical to the one obtained from liquid initialization, as is evident from their matching $\max(S_q)$ values and energies (within error bars). In contrast, at $B = 0$, we use different k -space meshes (see Sec. III B) for the liquid and crystal ansatz, so the crystal ansatz cannot flow to a liquid state through optimization. $N = 64$ in (a,b) and $N = 121$ in (c,d). As always, we assume full spin polarization. FL = Fermi liquid, WC = Wigner crystal.

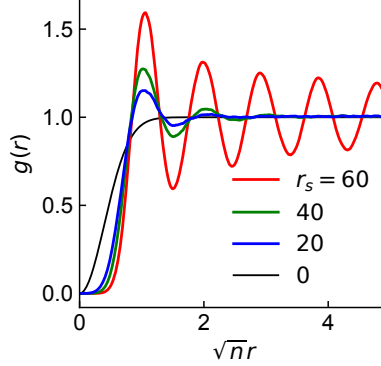


FIG. 2. **Pair correlation function.** Rotation-averaged pair correlation function of the $\nu = 1$ ground states at several interaction strengths. The curve at $r_s = 0$ is computed analytically and the curves at $r_s > 0$ are taken from sampling optimized wavefunctions (integer quantum Hall ansatz at $r_s = 20, 40$ and Landau level basis expansion ansatz at $r_s = 60$). $N=144$.

B. Pair correlation function

In Fig. 2, we show rotation-averaged pair correlation functions of the ground state at $\nu = 1$ for several r_s values.

C. Exact diagonalization benchmark for small system size

In Fig. 3, we compare energies from VMC, Hartree-Fock, and exact diagonalization on a small system with $N = 4$ electrons. We show the energies of both the full lowest Landau level Slater determinant and self-consistent Hartree-Fock ground state, which is a crystal for the r_s values shown. For the VMC wavefunction, we find that the integer quantum Hall orbitals are optimal for all r_s studied. This is sensible given that the system is extremely small, blurring the boundary between short and long-range order. Fig. 3 (b) shows that the VMC wavefunction covers the vast majority of the correlation energy.

In Fig. 4, we show variational energies for systems with $N = 9, 36, 121$ and 144 electrons for integer quantum Hall, Landau level basis expansion, and Gaussian orbitals. Generally, we find that the Gaussian orbitals perform similarly to the LL basis expansion orbitals. On the larger system sizes $N = 121$ and 144 , the Gaussian orbitals outperform the Landau level basis expanded orbitals

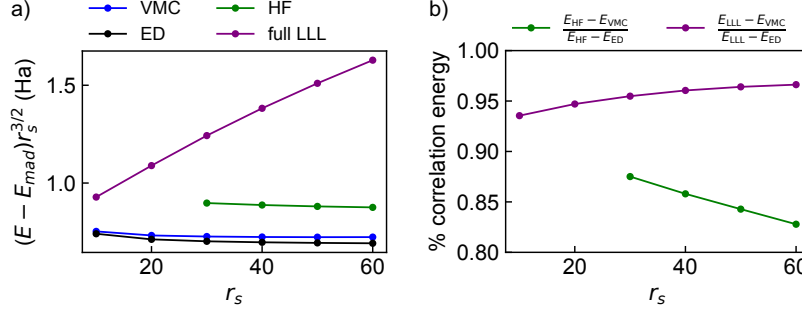


FIG. 3. **Benchmark with exact diagonalization for 4 electrons.** (a) Comparison of energies from self-consistent Hartree-Fock (HF), the full lowest Landau level Slater determinant, exact diagonalization (ED), variational Monte Carlo (VMC) wavefunctions. The exact diagonalization calculation includes 15 Landau levels. We believe that the exact diagonalization energy is converged with respect to Landau level truncation because the weight of the $n = 10$ through 14 Landau levels all together variational ground state is less than 5×10^{-4} at $r_s = 60$. (b) Percentage of the correlation energy recovered by the VMC wavefunction. We consider two definitions of the correlation energy: one where the ‘‘Hartree-Fock’’ energy is defined with respect to the full lowest Landau level Slater determinant and one where it is defined with respect to the self-consistent ground state. In both cases, the percentage of the correlation energy recovered by the VMC wavefunction is high, demonstrating its accuracy.

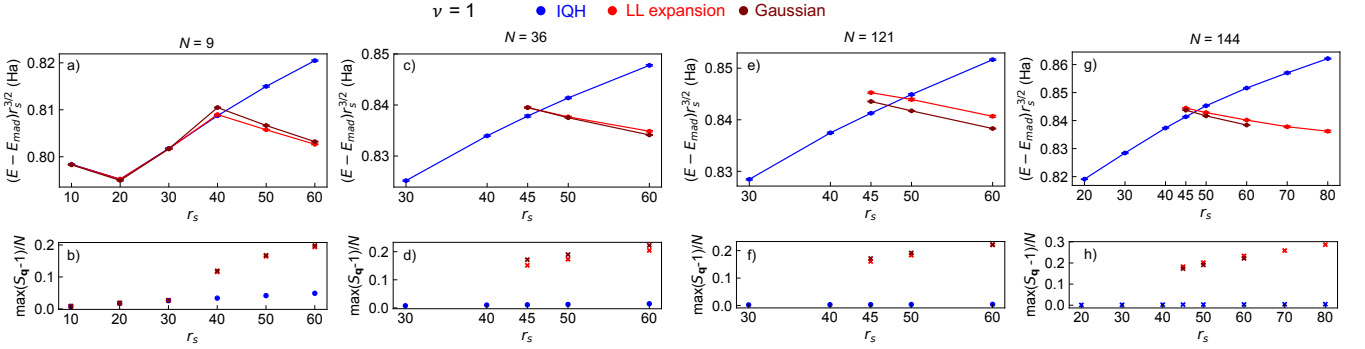


FIG. 4. **Comparison of $\nu = 1$ variational energies for multiple ansatzes and system sizes.** Energy per particle for various system sizes (top row) and magnitude of structure factor peaks. The values of r_s for the liquid transition we estimate by linearly interpolating the energies for $N = 9, 36, 121$ and 144 are $\approx 40, 46.5, 47$, and 47 . IQH = integer quantum Hall

slightly. However, the consistency in the estimates of the critical r_s across $N = 36, N = 121$, and $N = 144$ supports their accuracy.

D. Table of variational energies

TABLE I: **Table of energies of optimized variational wavefunctions.** N is the number of electrons. All energies are per particle. $E_{\text{mad}} = -1.106103 \text{ Ha}/r_s$ is the Madelung energy. Standard error is determined through binning analysis of local energy Monte Carlo data.

B field	Wavefunction type	N	r_s	$(E - E_{\text{mad}})r_s^{3/2} \text{ (Ha)}$	std. err.
$\nu = 1$	Integer quantum Hall	9	10	0.79841	6e-5
			20	0.7952	1.1e-4
			30	0.80179	9e-5
			40	0.80874	9e-5
			50	0.8150	1.4e-4
			60	0.8205	1.4e-4
		36	30	0.8252	1.2e-4
			40	0.8340	1.3e-4
			45	0.8378	1.9e-4
			50	0.8414	1.4e-4
			60	0.8478	1.5e-4
		121	30	0.8284	1.3e-4
			40	0.8375	1.2e-4
			45	0.8413	1.1e-4
			50	0.8449	1.4e-4
			60	0.8517	1.3e-4
		144	20	0.81909	7e-5
			30	0.8284	1.2e-4
			40	0.83736	9e-5
			45	0.8414	1.4e-4
			50	0.8453	1.3e-4
			60	0.8516	1e-4
			70	0.8570	2.6e-4
			80	0.8621	1.9e-4
	Landau level expansion	9	10	0.79834	6e-5
			20	0.79524	8e-5
			30	0.8017	1.3e-4
			40	0.8090	1.1e-4
			50	0.80574	7e-5
			60	0.80266	8e-5
		36	45	0.8395	1e-4
			50	0.8377	1e-4
			60	0.8349	1.4e-4
		121	45	0.8453	1.3e-4
			50	0.8439	2.1e-4
			60	0.8407	1.8e-4
		144	45	0.84448	9e-5
			50	0.8429	1.9e-4
			60	0.8402	1.7e-4
			70	0.8378	2.9e-4
			80	0.8362	4e-4
	Gaussian	9	10	0.79828	9e-5
			20	0.7950	1.2e-4
			30	0.8017	1.1e-4
			40	0.81048	8e-5
			50	0.8067	1.1e-4
			60	0.80322	8e-5
		36	45	0.8395	1e-4
			50	0.83751	8e-5
			60	0.83413	9e-5
		121	45	0.84355	7e-5
			50	0.84173	5e-5
			60	0.83830	8e-5
		144	45	0.84373	9e-5
			50	0.8417	1.1e-4

TABLE I: **Table of energies of optimized variational wavefunctions.** N is the number of electrons. All energies are per particle. $E_{\text{mad}} = -1.106103 \text{ Ha}/r_s$ is the Madelung energy. Standard error is determined through binning analysis of local energy Monte Carlo data.

B field	Wavefunction type	N	r_s	$(E - E_{\text{mad}})r_s^{3/2} \text{ (Ha)}$	std. err.
			60	0.83841	6e-5
$\nu = 2$	Integer quantum Hall	64	30	0.82384	7e-5
			35	0.82731	9e-5
			40	0.8307	1.2e-4
			45	0.83389	8e-5
	Landau level expansion	64	30	0.82376	8e-5
			35	0.8296	1.2e-4
			40	0.82896	9e-5
			45	0.8285	1e-4
	Gaussian	64	30	0.83043	6e-5
			35	0.8296	1e-4
			40	0.82876	8e-5
			45	0.82808	9e-5
$B = 0$	Fermi liquid	121	10	0.82401	2.2e-5
			20	0.81714	3e-5
			30	0.82260	4e-5
			40	0.82910	5e-5
			50	0.83524	7e-5
	Planewave expansion	121	10	0.83179	2.1e-5
			20	0.82154	3e-5
			30	0.82325	6e-5
			40	0.82462	6e-5
			50	0.8251	1e-4
	Gaussian	121	10	0.83191	2.8e-5
			20	0.82256	4e-5
			30	0.82378	4e-5
			40	0.82434	5e-5
			50	0.82417	5e-5

II. HARTREE-FOCK AND LINDEMANN CRITERION ANALYSIS

A. Hartree-Fock phase diagram and VMC energy comparison

Here we study the quantum phase diagram of the spin-polarized 2DEG at integral ν within the Hartree-Fock approximation. We adopt the same boundary conditions and supercell geometry as in our variational Monte Carlo calculations (see Sec. III C). We include 14 Landau levels and use supercells containing ≥ 200 flux quanta for every ν . We compute Landau level basis Coulomb matrix elements following the Supplementary Material of Ref. [4].

We compare the variational energies obtained by self-consistent iteration starting from three different density matrices: the non-interacting ground state density matrix, a random initial density matrix, and a Wigner crystal ‘‘seed’’ density matrix. In all cases, the self-consistent solutions obtained from random initial density matrices have energy higher than or equal to the energies obtained from the other initial density matrices. In the presence of interactions, the non-interacting integer quantum Hall ground state of ν full Landau levels is not a perfectly self-consistent solution at finite system size in the presence of interactions. However, we find numerically that it is self-consistent to an excellent approximation at the system sizes used here and we believe it is perfectly self-consistent in the thermodynamic limit.

In Fig. 5, we show the Hartree-Fock quantum phase diagram. For each ν , we perform calculations on a regular grid of r_s values with spacing 0.5. We estimate the transition points $r_{s,\nu}^*$ (black dots) by making a linear approximation to the energies of the liquid and crystal phases according to their values at the two r_s points closest to the transition. As shown in Fig. 5(a), $r_{s,\nu}^*$ monotonically decreases as ν increases, approaching the $B = 0$ limit of ≈ 2.7 as ν increases [5].

In Fig. 6, we compare Hartree-Fock and variational Monte Carlo energies at $\nu = 1$. Hartree-Fock is strongly biased in favor of the Wigner crystal. The large difference between the energies of the full lowest Landau Level Slater determinant (black) and the variational Monte Carlo liquid state (blue) energies demonstrates significant correlations in the liquid phase.

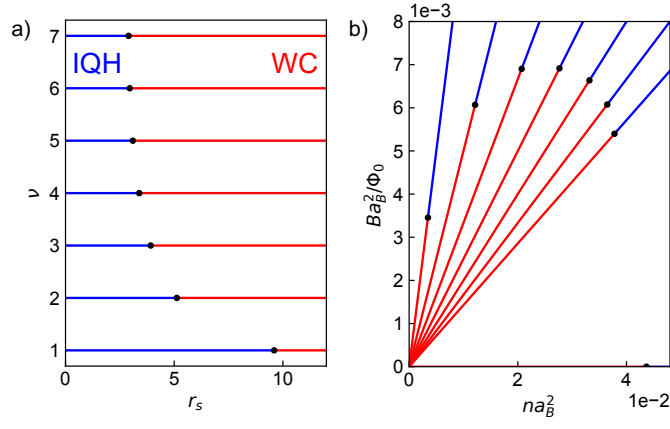


FIG. 5. Hartree-Fock quantum phase diagrams of the fully spin-polarized 2DEG at several integral Landau level filling factors ν . (a) and n - B plane (b). For each ν , there are two phases, integer quantum Hall (blue) and Wigner crystal (red), separated by a first-order phase transitions (black dots).

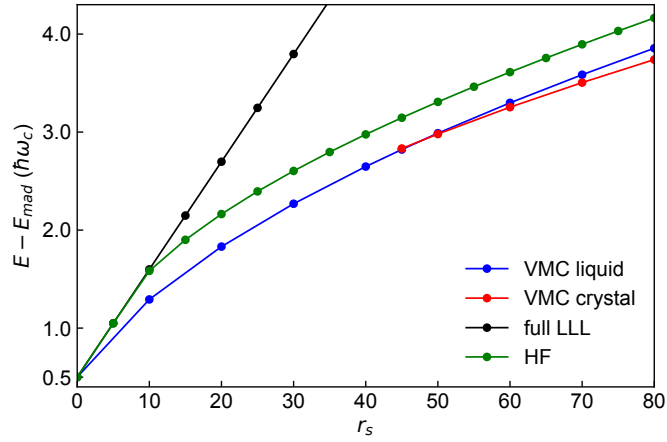


FIG. 6. Comparison of Hartree-Fock and variational Monte Carlo energies at $\nu = 1$. The ‘‘full LLL’’ energy is that of the full lowest Landau Level Slater determinant (the non-interacting ground state). The self-consistent HF energy is the energy of the lowest self-consistent HF state, which is a crystal for $r_s \gtrsim 9.6$ (see Fig. 5). VMC data are the same as in Fig. (1) of the main text and we omit statistical error bars here. $N = 144$.

B. Liquid energy in the Hartree-Fock approximation

To produce the liquid energy curve in Fig. 1(a) of the main text, we compute the energy expectation value of the thermal density matrix ρ of a non-interacting system in the limit $T \rightarrow 0$. In this density matrix, all degenerate many-body ground states of the kinetic energy operator are equally occupied and all higher energy states are unoccupied. The kinetic energy per particle of this density matrix is

$$K = \frac{\hbar \omega_c}{\nu} \left[\frac{[\nu]^2}{2} + (\nu - [\nu]) \left([\nu] + \frac{1}{2} \right) \right]. \quad (1)$$

The exchange energy per particle is

$$\Sigma_X = -\frac{e^2}{\ell} \frac{1}{2\nu} \sum_{m,n} \nu_n \nu_m X_{nm} \quad (2)$$

where

$$X_{nm} = \frac{\min(n, m)!}{\max(n, m)!} \int_0^\infty dx e^{-x^2/2} \left(\frac{x^2}{2} \right)^{|m-n|} \left[L_{\min(n, m)}^{[m-n]} \left(\frac{x^2}{2} \right) \right]^2 \quad (3)$$

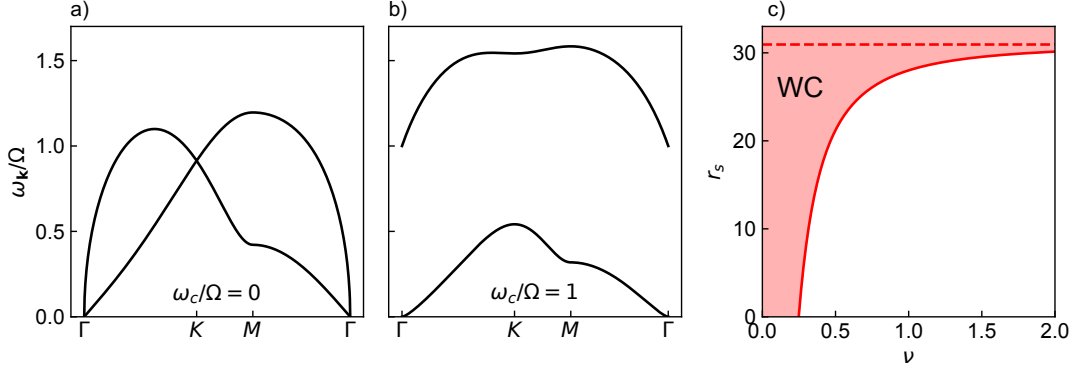


FIG. 7. **Phonons and Lindemann melting criterion.** (a,b) Phonon spectra in the harmonic approximation at (a) $\omega_c = 0$ and (b) $\omega_c = \Omega \equiv \sqrt{\frac{8e^2}{md_{\text{WC}}^3}}$. (c) Melting curve of the Wigner crystal in a magnetic field according to the Lindemann criterion with Lindemann ratio $\gamma^* = 0.3$. The pink region is the crystal phase, the solid red line is the melting curve, and the red dashed line marks the melting point at $B = 0$ (equivalently, $\nu = \infty$).

and ν_n is the occupation of the n^{th} Landau level ($\nu_n = 1$ if $n < \lfloor \nu \rfloor$ and $= \max(0, \nu - n)$ otherwise). L_n^m is a generalized Laguerre polynomial. We derived this expression following the discussion of Landau level basis Coulomb matrix elements in the Supplementary Material of Ref. [4].

At integral ν , K takes the same value for all ν while Σ_X tends downward as ν decreases. Needless to say, this infinitesimal-temperature Hartree-Fock approximation neglects many features of the liquid energy, particularly at non-integral ν . However, it correctly captures a robust feature of the liquid energy that we emphasize in the introduction of the main text: quantum oscillations with cusp minima at integral filling factors.

C. Phonons and Lindemann melting curve

Here we apply the heuristic quantum Lindemann criterion to study the boundary of the Wigner crystal phase in a magnetic field. The Lindemann criterion asserts that when the Lindemann ratio $\gamma = \sqrt{\langle \mathbf{u}^2 \rangle} / a_{\text{WC}}$ exceeds a critical value γ^* where $\mathbf{u}$ is the deviation of an electron from its equilibrium position and a_{WC} is the lattice constant, the crystal melts. Following Ref. [6], we compute the quantum ($T = 0$) uncertainty in the position of an electron in a Wigner crystal subjected to a magnetic field within the harmonic approximation as

$$\langle \mathbf{u}^2 \rangle = \frac{\hbar}{2m} \frac{1}{N} \sum_{\mathbf{k}} \frac{\omega_{1,\mathbf{k}} + \omega_{2,\mathbf{k}}}{(\omega_{1,\mathbf{k}} \omega_{2,\mathbf{k}})^2} \frac{1}{\sqrt{\omega_c^2 + (\omega_{1,\mathbf{k}} + \omega_{2,\mathbf{k}})^2}}. \quad (4)$$

Here, $\omega_{1,\mathbf{k}}$ and $\omega_{2,\mathbf{k}}$ are the Wigner crystal's longitudinal and transverse phonon frequencies at $B = 0$, which we calculate following Ref. [7]. We show the phonon dispersion at both zero and nonzero magnetic field strength in Fig. 7(a,b). In Fig. 7(c), we show the Lindemann melting curve according to Eq. 4. By inspection, Eq. 4 implies that $\langle \mathbf{u}^2 \rangle$ monotonically decreases as a function of $\omega_c \propto B$ and, therefore, that increasing B never melts a Wigner crystal.

This effect can be understood by approximating each electron as an uncoupled oscillator in a magnetic field and a harmonic potential coming from its interactions with all other electrons. In this approximation, the wavefunction of each electron is a Gaussian $\propto \exp\left(-\frac{m\sqrt{\omega_0 + (\frac{eB}{2mc})^2}}{2\hbar} r^2\right)$ whose position uncertainty strictly decreases as a function of B (ω_0 is the frequency of the Einstein oscillator at $B = 0$).

Ultimately, the Lindemann criterion fails to capture quantum melting of a $B = 0$ Wigner crystal into an integer quantum Hall state because it considers the density distribution of the crystal state rather than the energies of the liquid and crystal states, which ultimately dictate the phase boundary.

III. EXTENDED METHODS

A. Magnetic Bloch orbitals

Here we discuss magnetic Bloch orbitals, which have previously been discussed elsewhere [4, 8, 9]. These orbitals comprise a complete basis for each Landau level and simultaneously diagonalize magnetic translations on a magnetic lattice enclosing one flux quantum per unit cell.

We consider the Hamiltonian of a single electron with charge $-e$ confined to a two-dimensional plane in a homogeneous, out-of-plane magnetic field,

$$H = \frac{\pi^2}{2m} \quad (5)$$

with $\pi = \mathbf{p} + \frac{e}{c}\mathbf{A}(\mathbf{r})$. We set $\mathbf{B}(\mathbf{r}) = \nabla \times \mathbf{A} = -B\hat{z}$. Defining the ladder operators

$$\begin{aligned} a &= \frac{\ell}{\hbar\sqrt{2}} (\pi_x + i\pi_y) \\ a^\dagger &= \frac{\ell}{\hbar\sqrt{2}} (\pi_x - i\pi_y), \end{aligned} \quad (6)$$

which obey the algebra

$$[a, a^\dagger] = 1, \quad (7)$$

we have

$$H = \hbar\omega_c \left(a^\dagger a + \frac{1}{2} \right). \quad (8)$$

The condition $2\pi\ell^2 B = \Phi_0 = \frac{hc}{e}$ defines the magnetic length ℓ . The magnetic translation operator is

$$t(\mathbf{d}) = e^{i\chi\mathbf{d}(\mathbf{r})} T(\mathbf{d}), \quad (9)$$

where $\nabla\chi\mathbf{d}(\mathbf{r}) = \frac{2\pi}{\Phi_0} [\mathbf{A}(\mathbf{r} + \mathbf{d}) - \mathbf{A}(\mathbf{r})]$. $T(\mathbf{d}) = e^{i\mathbf{p}\cdot\mathbf{d}/\hbar}$ is an ordinary translation operator. Magnetic translations obey

$$[t(\mathbf{d}), \pi] = 0 = [t(\mathbf{d}), H] \quad (10)$$

and

$$t(\mathbf{d}')t(\mathbf{d}) = e^{i\mathbf{d}'\wedge\mathbf{d}/(2\ell^2)} t(\mathbf{d}' + \mathbf{d}) = e^{i\mathbf{d}'\wedge\mathbf{d}/\ell^2} t(\mathbf{d})t(\mathbf{d}'). \quad (11)$$

We also define the guiding center operator $\mathbf{R} = \mathbf{r} + \frac{\ell^2}{\hbar}\hat{z} \times \pi$, which obeys $[R_a, \pi_b] = 0$.

We seek an eigenbasis of H obeying the boundary condition

$$t(\mathbf{L}_i) |\phi\rangle = |\phi\rangle \quad (12)$$

where $\mathbf{L} = l_1\mathbf{L}_1 + l_2\mathbf{L}_2$ is a supercell lattice vector. $\mathbf{L}_{1,2}$ are primitive vectors of the supercell lattice. Let $N_\Phi = |\mathbf{L}_1 \times \mathbf{L}_2|/(2\pi\ell^2)$ and N_1, N_2 be integers such that $N_1N_2 = N_\Phi$. Let $\mathbf{a}_i = \mathbf{L}_i/N_i$ so that $|\mathbf{a}_1 \times \mathbf{a}_2| = 2\pi\ell^2$, implying $[t(\mathbf{a}_1), t(\mathbf{a}_2)] = 0$. $\mathbf{a}_{1,2}$ are primitive vectors of the magnetic lattice. We define the state $|f_0\rangle$ by the conditions $a|f_0\rangle = 0$ and $(R_x - iR_y)|f_0\rangle = 0$. We also define $|f_n\rangle = \frac{1}{\sqrt{n!}} a^\dagger |f_0\rangle$.

We seek to construct magnetic Bloch states $|\phi_{nk}\rangle$ that obey the eigenvalue equations

$$t(\mathbf{a}_i) |\phi_{nk}\rangle = e^{ik\cdot\mathbf{a}_i} |\phi_{nk}\rangle \quad (13)$$

and

$$a^\dagger a |\phi_{nk}\rangle = n |\phi_{nk}\rangle. \quad (14)$$

To proceed, we first define

$$\begin{aligned} |\phi_{n0}\rangle &= \sum_{r,s \in \mathbb{Z}} t(r\mathbf{a}_1)t(s\mathbf{a}_2) |f_n\rangle \\ &= \sum_{r,s \in \mathbb{Z}} e^{i\pi r s} t(r\mathbf{a}_1 + s\mathbf{a}_2) |f_n\rangle \end{aligned} \quad (15)$$

and then define

$$\begin{aligned} |\phi_{nk}\rangle &= t(\ell^2 \hat{z} \times \mathbf{k}) |\phi_{n0}\rangle \\ &= \sum_{r,s \in \mathbb{Z}} e^{-i\mathbf{k} \cdot (r\mathbf{a}_1 + s\mathbf{a}_2)} e^{i\pi r s} t(r\mathbf{a}_1 + s\mathbf{a}_2) t(\ell^2 \hat{z} \times \mathbf{k}) |f_n\rangle \end{aligned} \quad (16)$$

Given these definitions, Eq. 10 implies Eq. 14 and Eq. 11 implies Eq. 13.

To make Eqs. 12 and 13 compatible, $\mathbf{k}$ can take any value $\mathbf{k} = k_1 \mathbf{G}_1 + k_2 \mathbf{G}_2$ where $\mathbf{L}_i \cdot \mathbf{G}_j = 2\pi \delta_{ij}$. There are N_Φ unique values of $\mathbf{k}$, with integral $0 \leq k_i < N_i$, in the first magnetic Brillouin zone. The magnetic Bloch states thus comprise a complete basis of N_Φ states per Landau level.

For numerical calculations, we use the symmetric gauge $\mathbf{A}(\mathbf{r}) = \frac{B}{2}(+y, -x)$. In this gauge, $\chi_{\mathbf{d}}(\mathbf{r}) = \mathbf{r} \wedge \mathbf{d}/(2\ell^2)$,

$$f_n(\mathbf{r}) = \frac{1}{\ell \sqrt{2n!}} (i\bar{z})^n e^{-r^2/(4\ell^2)} \quad (17)$$

and

$$\phi_{nk}(\mathbf{r}) = \sum_{\mathbf{a}} e^{i(\frac{\mathbf{k} \cdot (\mathbf{r} - \mathbf{a})}{2} + \frac{\mathbf{r} \wedge \mathbf{a}}{2\ell^2} + \pi r s)} f_n(\mathbf{r} + \mathbf{a} + \ell^2 \hat{z} \times \mathbf{k}) \quad (18)$$

where $\mathbf{a} = r\mathbf{a}_1 + s\mathbf{a}_2$. This infinite sum over magnetic lattice vectors $\mathbf{a}$ converges rapidly due to the Gaussian factor in Eq. 17.

The identity

$$\phi_{nk}(\mathbf{r} + \mathbf{a}) = e^{i(\mathbf{k} \cdot (\mathbf{a} + \mathbf{r}/2) - \frac{\mathbf{r} \wedge \mathbf{a}}{2\ell^2} - \pi r s)} \phi_{n0}(\mathbf{r} - \ell^2 \mathbf{k} \times \hat{z}) \quad (19)$$

is useful for numerical calculation because it allows one to easily calculate the value of ϕ_{nk} at any position knowing only ϕ_{n0} in a single magnetic unit cell. Exploiting this formula, we precompute ϕ_{n0} on a fine grid (100×100 points) in a single magnetic unit cell and interpolate its real and imaginary parts with two-dimensional cubic B -spline interpolants on this grid [10]. We then combine this interpolation with Eq. 19 to quickly compute $\phi_{nk}(\mathbf{r})$.

We note that the algebraic relationship $\pi_x |\phi_{nk}\rangle = \frac{\hbar}{\ell \sqrt{2}} (a^\dagger + a) |\phi_{nk}\rangle = \frac{\hbar}{\ell \sqrt{2}} \left[\sqrt{n+1} |\phi_{(n+1)\mathbf{k}}\rangle + \sqrt{n} |\phi_{(n-1)\mathbf{k}}\rangle \right]$ (and similar for π_y) is useful for computing the local energy.

B. Plane wave expansion at $B = 0$

At $B = 0$, we impose periodic boundary conditions and use plane wave basis for the single-particle orbitals:

$$|\psi_{nk}\rangle = \sum_m U_{nm}^k |\mathbf{k} + \mathbf{g}_m\rangle \quad (20)$$

where $\langle \mathbf{r} | \mathbf{k} \rangle = e^{i\mathbf{k} \cdot \mathbf{r}}$. For the Fermi liquid ansatz, each electron occupies one of the N lowest-energy plane waves and there is no variational freedom in the orbitals. For $N = 121$, this set is unique, i.e. the system is ‘‘closed shell’’ For the Wigner crystal, we choose a Wigner-Seitz Brillouin zone mesh. For each point on this mesh, we expand a Bloch orbital in a plane wave basis out to the second shell of reciprocal lattice vectors with 13 reciprocal lattice vectors in total (see Fig. 8).

C. Supercell geometry

We use a hexagonal magnetic unit cell with $\mathbf{a}_1 = \nu a_{\text{WC}}(\frac{\sqrt{3}}{2}, -\frac{1}{2})$ and $\mathbf{a}_2 = a_{\text{WC}}(\frac{\sqrt{3}}{2}, +\frac{1}{2})$ where $a_{\text{WC}} = \sqrt{\frac{2}{\sqrt{3}n}}$. ν appears here so that the magnetic unit cell has an aspect ratio of ν and is commensurate with the Wigner crystal lattice. For a system with N electrons, we set $N_1 = \sqrt{N}/\nu$ and $N_2 = \sqrt{N}$ so that the supercell always has an equal aspect ratio. An example of a magnetic Brillouin zone mesh is shown in Fig. 8(a).

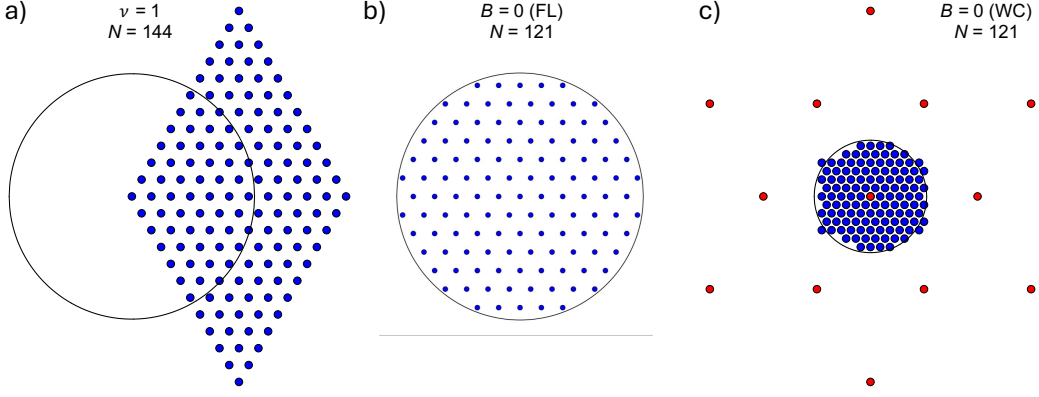


FIG. 8. Diagrams of momentum-space meshes used in VMC calculations. The black circle is centered about the origin and has radius $k_F = \sqrt{n/4\pi}$. The different panels are not to scale. Blue dots denote the mesh of $\mathbf{k}$ points and the red dots denote reciprocal lattice vectors $\mathbf{g}$, which are only relevant to the $B = 0$ Wigner crystal.

At $B = 0$, the same setup applies upon replacing $\nu \rightarrow 1$. Momentum space meshes for the $B = 0$ Fermi liquid and Wigner crystal ansatz are shown in Fig. 8(a,b). We remark that, at $B \neq 0$, all valid magnetic Brillouin zones give the same Hilbert space. In contrast, at $B = 0$, the choice of mesh clearly matters for the Fermi liquid phase and also matters for the Wigner crystal phase because of the finite reciprocal lattice truncation.

D. Gaussian orbitals

At $B = 0$, we define Gaussian orbitals obeying periodic boundary conditions as

$$\psi_{\mathbf{R}}(\mathbf{r}) = \frac{1}{\sqrt{2\pi}\ell_0} \sum_{\mathbf{L}} e^{-\frac{(\mathbf{r}-\mathbf{R}+\mathbf{L})^2}{4\ell_0^2}} \quad (21)$$

where $\mathbf{R}$ is the center of the Gaussian within the supercell, $\mathbf{L}$ are supercell lattice vectors, and ℓ_0 is a variational parameter.

At $B \neq 0$, we define Gaussian orbitals obeying magnetic boundary conditions as

$$\begin{aligned} |\psi_{\mathbf{R}}\rangle &= \sum_{n,m \in \mathbb{Z}} t(n\mathbf{L}_1)t(m\mathbf{L}_2) |\phi\rangle \\ &= \sum_{\mathbf{L}} e^{i\pi N_{\Phi} n m} t(\mathbf{L}) |\phi\rangle \end{aligned} \quad (22)$$

where $\mathbf{L} = n\mathbf{L}_1 + m\mathbf{L}_2$. Here,

$$\phi(\mathbf{r}) = \langle \mathbf{r} | \phi \rangle = \frac{1}{\sqrt{2\pi}\ell_0} e^{-\frac{r^2}{4\ell_0^2}}. \quad (23)$$

In the symmetric gauge,

$$t(\mathbf{d}) = e^{i\frac{\mathbf{r} \wedge \mathbf{d}}{2\ell^2}} T(\mathbf{d}) \quad (24)$$

and

$$\psi_{\mathbf{R}}(\mathbf{r}) = \frac{1}{\sqrt{2\pi}\ell_0} \sum_{\mathbf{L}} e^{i\left(\pi N_{\Phi} n m + \frac{\mathbf{r} \wedge \mathbf{L}}{2\ell^2}\right)} e^{-\frac{(\mathbf{r}-\mathbf{R}+\mathbf{L})^2}{4\ell_0^2}}. \quad (25)$$

E. Jastrow factor

Our two-body Jastrow factor takes the form

$$J(R) = \exp \left(\sum_{i < j} u(\mathbf{r}_i - \mathbf{r}_j) \right) \quad (26)$$

where

$$u(\mathbf{r}) = S(f(\mathbf{r})), \quad (27)$$

$$f(\mathbf{r}) = \frac{2}{3} \sqrt{\sum_{i=1,2,3} \sin^2(\mathbf{G}_i \cdot \mathbf{r}/2)}, \quad (28)$$

and

$$S(x) = \sum_{m=0}^M c_m B_3(Mx - m). \quad (29)$$

Here

$$B_3(x) = \begin{cases} \frac{1}{6} (4 - 6|x+1|^2 + 3|x+1|^3) & 0 \leq |x+1| \leq 1, \\ \frac{1}{6} (2 - |x+1|^3) & 1 \leq |x+1| \leq 2, \\ 0 & \text{otherwise} \end{cases} \quad (30)$$

is the cardinal cubic B -spline function, which is twice differentiable everywhere and nonzero only in the region $-3 \leq x \leq 1$. $f(\mathbf{r})$ is scaled so that $\max f(\mathbf{r}) = 1$ and is a ‘‘periodification’’ of $|\mathbf{r}|$ in the sense that (1) $f(\mathbf{r}) \propto |\mathbf{r}|$ at small $|\mathbf{r}|$ and (2) $f(\mathbf{r} + \mathbf{L}) = f(\mathbf{r})$ where $\mathbf{L}$ is a supercell lattice vector. Here $\mathbf{G}_1 = \frac{2\pi\hat{z}}{|\mathbf{L}_1 \times \mathbf{L}_2|}$, $\mathbf{G}_2 = \frac{2\pi\hat{x}}{|\mathbf{L}_1 \times \mathbf{L}_2|}$, and $\mathbf{G}_3 = \frac{\mathbf{G}_1 + \mathbf{G}_2}{2}$ are reciprocal lattice vectors of the supercell. $f(\mathbf{r})$ is approximately isotropic at small $|\mathbf{r}|$ for supercells with an equal aspect ratio, which we always use in this work.

Our parameterization of $u(\mathbf{r})$ is intended to capture both short and long-range correlations in a unified form. It is inspired by the two-body spine Jastrow in QMCPack [11] and the Jastrow factor described in the Supplementary Material of Ref. [12]. The spline function $S(x)$ is highly expressive.

The cusp condition (in this case, for parallel spins in two dimensions)

$$\hat{\mathbf{r}} \cdot \nabla u(\mathbf{r}) = \frac{1}{3a_B} \quad (31)$$

eliminates divergence in the local energy where two particles coincide [13, 14]. It imposes the constraint $c_0 = c_2 - \frac{1}{\pi\sqrt{2}M} \frac{|\mathbf{L}_1|}{a_B}$. c_1 through c_M are optimizable parameters and c_0 is fixed by the cusp condition. The number M of ‘‘control points’’ is a free hyperparameter that we set between 8 and 15.

F. Local energy

The local energy is defined as

$$E_L(R) = \frac{\langle R | H | \Psi \rangle}{\langle R | \Psi \rangle} = \frac{1}{\Psi(R)} H \Psi(R). \quad (32)$$

The kinetic contribution to the local energy of our Slater-Jastrow wavefunction,

$$\Psi(R) = \det [\psi_i(\mathbf{r}_j)] \exp \left(\sum_{i < j} u(\mathbf{r}_i - \mathbf{r}_j) \right) \equiv D(R) e^{U(R)}, \quad (33)$$

is

$$\begin{aligned}
T_L &= \frac{1}{2m} \sum_i \frac{\pi_i^2 \Psi}{\Psi} \\
&= \frac{1}{2m} \sum_i \left[\frac{\pi_i^2 D}{D} - \hbar^2 \nabla_i^2 U - (\hbar \nabla_i U)^2 - 2i\hbar \nabla_i U \cdot \frac{\pi_i D}{D} \right]
\end{aligned} \tag{34}$$

where we recall that $\pi = -i\hbar \nabla + \frac{e}{c} \mathbf{A}(\mathbf{r})$ and we have suppressed explicit R -dependence. We compute the Coulomb potential contribution using the Ewald method (see, for example, Ref. [15]).

G. Orbital parameterization

Our approach to orbital parameterization is inspired by that of Ref. [16]. We parameterize our single-particle orbitals as

$$|\psi_{\mathbf{k}m}\rangle = \sum_{n=0}^{n_{\max}} U_{mn}^{\mathbf{k}} |\phi_{\mathbf{k}n}\rangle \tag{35}$$

where

$$U^{\mathbf{k}} = \exp(i\Theta^{\mathbf{k}}) \tag{36}$$

is the unitary orbital rotation matrix and $\Theta_{mn}^{\mathbf{k}}$ is a Hermitian matrix. At filling factor $\nu \in \mathbb{Z}$, ν orbitals are occupied and $(n_{\max}+1-\nu)$ are unoccupied at each $\mathbf{k}$. The matrix elements $\Theta_{mn}^{\mathbf{k}}$, where m and n are both in the occupied sector ($< \nu$) or both in the unoccupied sector ($\geq \nu$), generate unitary transformations within the occupied or unoccupied sector whose only effect is to multiply the many-body wavefunction by an overall phase. Therefore, we set them to 0. We are left with $(\# \text{ of } \mathbf{k} \text{ points}) \times \nu \times (n_{\max} + 1 - \nu) \times 2$ orbital rotation parameters: $\theta_{mn}^{\mathbf{k},R}$ and $\theta_{mn}^{\mathbf{k},I}$ for $0 \leq m < \nu$ and $\nu \leq n \leq n_{\max} - 1$ where

$$\Theta_{mn}^{\mathbf{k}} = \begin{cases} \theta_{mn}^{\mathbf{k},R} + i\theta_{mn}^{\mathbf{k},I} & \text{if } m < n \\ \theta_{nm}^{\mathbf{k},R} - i\theta_{nm}^{\mathbf{k},I} & \text{if } n < m. \end{cases} \tag{37}$$

To run stochastic reconfiguration (see Ref. [17]), we need the quantity $\partial_\alpha \ln \Psi(R)$ where α is a variational parameter. For orbital parameters, we use the formula

$$\partial_\alpha \ln \Psi(R) = \frac{\partial_\alpha \det \psi_i(\mathbf{r}_j)}{\det \psi_i(\mathbf{r}_j)} = \frac{\det \psi_i^\alpha(\mathbf{r}_j)}{\det \psi_i(\mathbf{r}_j)} \tag{38}$$

where

$$\psi_i^\alpha(\mathbf{r}_j) = \begin{cases} \partial_\alpha \psi_i(\mathbf{r}_j) & \text{if } \alpha \text{ and } i \text{ correspond to the same } \mathbf{k} \\ \psi_i(\mathbf{r}_j) & \text{else.} \end{cases} \tag{39}$$

Here $i \sim m\mathbf{k}$ is a compound orbital index. From here on we will drop the explicit $\mathbf{k}$ index because all terms will share it. We now need formulae for

$$\frac{\partial |\psi_m\rangle}{\partial \theta_{pq}^R} = \left(\frac{\partial}{\partial \Theta_{pq}} + \frac{\partial}{\partial \Theta_{qp}} \right) |\psi_m\rangle \tag{40}$$

and

$$\frac{\partial |\psi_m\rangle}{\partial \theta_{pq}^I} = i \left(\frac{\partial}{\partial \Theta_{pq}} - \frac{\partial}{\partial \Theta_{qp}} \right) |\psi_m\rangle. \tag{41}$$

First, we have

$$\begin{aligned}
\frac{\partial |\psi_m\rangle}{\partial \Theta_{pq}} &= \frac{\partial}{\partial \Theta_{pq}} \sum_n U_{mn} |\phi_n\rangle \\
&= \sum_n \frac{\partial \exp(i\Theta)_{mn}}{\partial \Theta_{pq}} |\phi_n\rangle
\end{aligned} \tag{42}$$

where the $|\phi\rangle'$ s are basis orbitals. Finally, from Theorem 4.5 of Ref. [18], it follows that

$$\frac{\partial \exp(i\Theta)_{mn}}{\partial \Theta_{pq}} = i \sum_{r,s} V_{mr} V_{rp}^\dagger V_{qs} \Phi_{rs} V_{sn}^\dagger \quad (43)$$

where $V\Lambda V^\dagger = i\Theta$ is the spectral decomposition of $i\Theta$ and

$$\Phi_{pq} = \begin{cases} \frac{e^{\Lambda_{pp}} - e^{\Lambda_{qq}}}{\Lambda_{pp} - \Lambda_{qq}} & \text{if } p \neq q \\ e^{\Lambda_{pp}} & \text{if } p = q. \end{cases} \quad (44)$$

These are all the formulae we need to implement our orbital parameterization and its optimization via stochastic reconfiguration.

H. Sampling

We sample the probability distribution $|\Psi(R)|^2$ using the standard Metropolis-Hastings algorithm, proposing single-particle moves from a Gaussian distribution whose radius is updated between optimization steps to maintain a 50% acceptance rate [19]. We use the method of Ref. [20] to efficiently compute ratios of determinants of Slater matrices that differ only in one column.

During wavefunction optimization, we typically take of order 10^4 samples per optimization step. At later stages of optimization, we take up to 10^6 per optimization step. We sample the wavefunction after every single particle move. Once optimized, we typically take 10^7 samples to estimate energies and correlation functions. The autocorrelation length we find for the local energy of our optimized wavefunction is typically a few hundred to a few thousand steps.

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